

Xi Number-Space Dependence

Doc. 253 — ξ and Number-Space Dependence in Factorisation

Abstract

Working document. Purpose: preservation of results from an extended investigation without breakthrough. The question investigated was whether the FFGFT parameter $\xi = 4/30000$ exhibits number-space dependence in the factorisation of composite integers, and whether this dependence follows an FFGFT-type step hierarchy. All calculations were verified with Python. The connection to FFGFT remains a conjecture without an algebraic bridge proof.

1. Background and Motivation

During work on a T0/FFGFT-based factorisation approach (Python scripts and HTML test programs, 2024–2026), the following was observed: the effectiveness of a global ξ -parameter in period-finding (Shor's algorithm) does not appear constant, but depends on the number space. At the same time, rounding errors and computational precision limits enter the results indirectly. The observed dependence of the effectiveness of ξ on the number space is not a finding: in the implementations it was a hard-wired case distinction by bit length whose branches behave identically; there is no ξ -factorisation method distinguishable from Shor's (cf. Section 4).

This gave rise to two questions:

1. Is there a genuine number-space dependence of ξ in factorisation?
2. Does this dependence follow an FFGFT-type step hierarchy?

Status after extended work without breakthrough: Both questions are neither proved nor disproved. The calculations below provide the established facts.

2. The Algebraically Correct Quantity: $\xi_{\text{num}}(N)$

The HTML file `t0_xi_num_algebraisch.html` correctly defined the relevant quantity:

$$\xi_{\text{num}}(N) = \frac{\lambda(N)}{N}$$

where $\lambda(N) = \text{lcm}(p-1, q-1)$ is the Carmichael function for $N = p \cdot q$. This quantity is dimensionless and basis-independent.

Analytical approximation for large primes p, q :

$$\xi_{\text{num}}(N) = \frac{\text{lcm}(p-1, q-1)}{p \cdot q} \approx \frac{1}{\text{gcd}(p-1, q-1)}$$

Since $\text{lcm}(p-1, q-1) = (p-1)(q-1)/\text{gcd}(p-1, q-1)$, one has $\lambda/(pq) \approx 1/\text{gcd}$; confirmed by Table 3.1: 10403 ($\text{gcd} = 2$) $\rightarrow$ 0.4902, 9797 ($\text{gcd} = 4$) $\rightarrow$ 0.2450.

Key finding (from `t0_primzahl_periodizitaet.html`, Python-confirmed): The period structure of N is determined entirely by $\text{gcd}(p-1, q-1)$. This is a property of the primes themselves — not of a global parameter.

3. Python Verification: ξ_{num} Tables

3.1 Concrete Semiprimes

N	p	q	$\text{gcd}(p-1, q-1)$	$\lambda(N)$	ξ_{num}	Type
15	3	5	2	4	0.2667	Shor classic
35	5	7	2	12	0.3429	
143	11	13	2	60	0.4196	twin primes
65	5	13	4	12	0.1846	$\text{gcd} = 4$
301	7	43	6	42	0.1395	$\text{gcd} = 6$
77	7	11	2	30	0.3896	$\text{gcd} = 2$
4087	61	67	6	660	0.1615	twin ~ 60
9797	97	101	4	2400	0.2450	smooth $p-1$
10403	101	103	2	5100	0.4902	twin ~ 100
291	3	97	2	96	0.3299	very unequal

ξ_{num} lies in the range $[0.14, 0.49]$.
 $\xi_{\text{FFGFT}} = 1.333 \times 10^{-4}$ lies far outside this range.

3.2 Dependence on $\gcd(p-1, q-1)$

gcd	ξ_{num} (analytical)	Factorisability	Example
2	≈ 0.5	hard (RSA class)	twin primes
4	≈ 0.25	moderate	$p \equiv 1, q \equiv 1 \pmod{4}$
6	≈ 0.167	easier	$p \equiv 1, q \equiv 1 \pmod{6}$
12	≈ 0.083	easy	smooth $p-1, q-1$
30	≈ 0.033	very easy	B -smooth primes
60	≈ 0.017	trivial	highly smooth primes

The number-space dependence is real — but it is classical number theory. Pollard's p -method and Lenstra's ECM exploit precisely this gcd-structure. It is not an FFGFT-specific phenomenon.

4. What the HTML Tests Established

The following HTML test programs were developed: `t0_shor_hypothesis.html`, `t0_xi_num_algebraisch.html`, `t0_primzahl_periodizitaet.html`, `t0_shor_bigint.html`. They formulated the questions correctly but produced no systematic result tables.

Formulated hypotheses (now numerically clarified):

H_0 ξ -optimal bases and random bases produce periods of the same order of magnitude (no advantage from ξ -selection).

H_1 ξ -bases produce smaller periods in the median (factor > 1.5 over $N = 50$ trials).

The ξ -resonance function of the HTML tools ($\omega = 2\pi/r$, resonance = $\exp(-(\omega - \pi)^2/(4|\xi|))$); the Python library instead uses the Lorentzian form $1/(1 + |(\omega - \pi)^2/(4\xi)|)$ with the same maximum) is a heuristic without physical derivation. Since $\omega - \pi = \pi(2/r - 1)$, it is maximal at $r = 2$ for every ξ : it selects no resonance, only the smallest even order — which is the most common gcd-case, not a ξ -effect.

5. Relationship to ξ_{FFGFT}

$$\xi_{\text{FFGFT}} = \frac{4}{30000} = 1.333 \times 10^{-4} \ll \xi_{\text{num}} \in [0.017, 0.49]$$

(lower bound $1/60 = 0.017$ from Table 3.2).

The two quantities are not numerically identical and not simply related by a scale factor. A structural connection — for instance that ξ_{num} is a “projection” of ξ_{FFGFT} onto the number space — remains a conjecture without an algebraic bridge proof.

What is missing for a proof:

1. An argument for why ξ_{FFGFT} (geometric) should appear in ξ_{num} (arithmetic)
2. A scale relation explaining how $\xi_{\text{FFGFT}} \approx 10^{-4}$ maps onto $\xi_{\text{num}} \approx 0.1\text{--}0.5$
3. A prediction that differs from classical number theory

6. Established Facts — Summary

Statement	Status
$\xi_{\text{num}}(N)$ is number-space dependent	✓ Proved
The dependence is classical number theory	✓ Proved
No global ξ optimises factorisation universally	✓ Proved
ξ -resonance heuristic is maximal at $r = 2$ for every ξ	✓ Clear
Rounding errors affect effective ξ	✓ Known
ξ_{num} follows FFGFT step hierarchy	× Not proved
ξ_{FFGFT} and ξ_{num} structurally related	× Not proved
ξ -basis selection improves period-finding	× H_0 not refuted

Conclusion: The number-space dependence is real but classical. The connection to FFGFT geometry is open and remains a conjecture. The breakthrough is missing because the algebraic bridge proof is missing — it would need to show why a geometric invariant of T^4 appears in the Carmichael structure of arithmetic number spaces.